MAM · Integration and Its Applications — Patterns and Solution Approaches

Basis: marking keys + exam reports for a total of 39 past exam questions containing integration, 2016–2025. Original questions in `topics/02_Integration.pdf`. Integration is mainly Calculator-free (when it is the main knowledge point: CF 16 questions, CA 7 questions). Also cites 2 questions from the 2016 sample paper (`2016S-F1`, `2016S-A20`; original paper in `papers/2016Sample_*`). Review status: has been through one round of independent review and revised in line with feedback (see `Review_log.md`).

First, remember: where the marking panel repeatedly deducts marks

- FTC (Fundamental Theorem of Calculus) has repeatedly been called out in reports (2016, 2017, 2018, 2022, 2025 reports): not recognising the different uses of the FTC; the 2022 report specifically calls out the FTC combined with the chain rule (when the upper limit is g(x), multiply by g′(x)).
- Area ≠ integral: parts below the x-axis must be taken as absolute value, or subtracted piecewise (all 4 marks of 2018A-Q16(b) are here).
- dx must be written after the integral sign (2023 report: marks deducted for notation; 2025F-Q5 awarded 1 mark for "uses correct notation including dx on all integrals").
- Areas whose upper/lower boundaries are made up of multiple functions must be split into multiple integrals (called out in the 2024 report).
- Indefinite integrals must include + c (the key sometimes gives 1 mark for this on its own).
- "Show that" — you must not skip steps.

Pattern 1: Basic indefinite integrals and finding f from f′

Recognition: "Determine ∫…dx", "If f′(x) = …, find f(x) given f(0) = −2".

Approach

1. Memorise the antiderivatives: $\int e^{kx}dx = e^{kx}/k$; $\int \cos kx\,dx = \sin kx / k$; $\int (ax+b)^n dx = (ax+b)^{n+1}/[a(n+1)]$ $(n \neq -1)$; $\int f'/f\,dx = \ln|f| + c$ (in exams f(x) > 0 generally, and the key usually writes ln f(x) + c).
2. Evaluating definite integrals (commonly 3 marks in CF): ✓ antiderivative ✓ substitute the upper and lower limits ✓ simplify to an exact value (2017F-Q6(a), 2021F-Q4(b), 2023F-Q1(b), 2023F-Q2(a)(iii)).
3. When the numerator is a multiple of the derivative of the denominator: first pull out the coefficient, e.g. $\int (3x+1)/(3x^2+2x+1)dx = \frac{1}{2}\int (6x+2)/(\ldots)dx = \frac{1}{2}\ln(\ldots) + c$. The key gives 1 mark specifically for "modifies the integrand so the numerator is the derivative of the denominator".
4. Finding f from f′: integrate and write + c → substitute the known point to find c → write the full f(x). When the answer involves ln, simplify using log laws (e.g. ln 32 − ln 4 = ln 8).

Mark allocation: ✓ correct antiderivative ✓ include the constant c ✓ substitute the point to find c ✓ write the final expression (simplified as required).

Past questions: 2018F-Q3(c)(d), 2019F-Q1(c), 2021F-Q1(c), 2021F-Q4(a), 2022F-Q1(b), 2024F-Q1(b), 2020A-Q10(b).

Pattern 2: "Hence" — use the derivative just found to find the integral ★ CF frequently tested

Recognition: (a) first find d/dx(x ln x) or d/dx(2x sin 3x) (2017F-Q8); (b) "Hence show that $\int \ln x\,dx = x\ln x - x + c$".

The three fixed steps (the key usually awards marks by these three steps: 2017F-Q8 and 2022F-Q6 are exactly 3 marks; 2018F-Q7 is split into 4 marks, 2023F-Q2 is combined into 2 marks)

1. Integrate both sides: $\int \frac{d}{dx}(x\ln x)\,dx = \int (\ln x + 1)dx$.
2. Apply the FTC to the left side: left side = x ln x (the antiderivative and derivative cancel).

3. Split the right side and rearrange: $x \ln x + c_1 = \int \ln x\,dx + x + c_2 \Rightarrow \int \ln x\,dx = x \ln x - x + c$ (the two constants are combined into c).

Common errors: writing the conclusion directly; forgetting + c; not splitting the right side using linearity.
Past questions: 2016F-Q2(b), 2017F-Q8(b), 2018F-Q7(b), 2022F-Q6(a)(ii), 2023F-Q2(c)(ii). This is usually followed by a further part that uses this result to find an expectation, definite integral or area (2022F-Q6(b)(ii), 2023F-Q2(c)(iii), 2018F-Q7(c)).

Pattern 3: Definite integral = signed area (reading values from a graph)

Recognition: a graph made up of line segments, semicircles or quarter-circles is given, and you are asked for $\int_0^{12} f(x)dx$, $\int (f(x) - 2)dx$, $\int f'(x)dx$.
Approach

1. Split the region into rectangles, triangles and sectors; above the x-axis is positive, below is negative.
2. $\int (f(x) - k)dx = \int f\,dx - k\cdot$(length of the interval).
3. $\int_a^b f'(x)dx = f(b) - f(a)$ (FTC; read the two function values directly from the graph).
4. To find α such that $\int_0^{\alpha} f = 0$: set the positive area above equal to the negative area below and set up an equation.

Mark allocation: ✓ split areas ✓ add with signs ✓ result (exact value, including π).
Past questions: 2016S-F1, 2016F-Q7, 2022F-Q2.

Pattern 4: FTC and functions defined by integrals ★ Frequently tested (appears in 2016–18, 2021–23, 2025), and repeatedly called out in reports

Recognition: $A(x) = \int_a^x f(t)dt$; $\frac{d}{dx} \int_a^{g(x)} f(t)dt$; $\frac{d}{dt} \int_t^3 f(x)dx$; "Sketch A(x)", "Is F(2π) positive or negative?".
Core formulas

- $\frac{d}{dx} \int_a^x f(t)dt = f(x)$ (the result must be written as a function of x);
- $\frac{d}{dx} \int_a^{g(x)} f(t)dt = f(g(x))\cdot g'(x)$ (2022A-Q15: $D(t) = \int_0^{\pi t}\ldots \Rightarrow D'(t) = \pi\cdot\sqrt{(1+3\cos^2(\pi t))}$);
- when the lower limit is the variable, swap the limits first: $\int_t^3 f = -\int_3^t f$, so the derivative is $-f(t)$ (2022F-Q1(c));
- $\int_a^b f'(x)dx = f(b) - f(a)$ (2018A-Q16(a), 2023F-Q2(a)(iii), 2016F-Q4(c)).
- Linearity + integral function: $\int_2^4 (f + 2)dx = F(4) - F(2) + 4$ (2023F-Q5(b)).

Rules for sketching A(x) (the integral function)

- A(a) = 0 (at the lower limit); mark the zeros of A;
- where f changes from positive to negative, A has a local maximum; where f changes from negative to positive, A has a local minimum;
- turning points of f correspond to points of inflection of A;
- whether A is increasing or decreasing is determined by the sign of f; mark the value of A at the endpoints.

Deciding whether F(x) is positive or negative (2025F-Q2): F(x) equals the signed area from 0 to x; look at whether the positive or negative area on the graph is larger. The reason must refer to the areas — you cannot just state the conclusion.
Past questions: 2016F-Q5, 2016S-A20, 2017A-Q15(a), 2018A-Q16, 2021F-Q4(c), 2022A-Q15, 2022F-Q1(c), 2023F-Q5(b)(c), 2025F-Q2.

Pattern 5: Area under a curve / between curves

Recognition: "Determine the (exact) area bounded by …", or "Write an integral expression for the area".

Approach

1. Find the intersection points first as the limits of integration (the intersection points themselves often earn 1 mark).
2. Decide which is on top: area = ∫(top − bottom)dx.
3. When finding the area between a curve and the x-axis and the curve crosses the x-axis, or when the top/bottom relationship between the two curves swaps, you must split into pieces (when finding only the area between two curves, a curve crossing the x-axis does not by itself require splitting).
4. When the area is not convenient to integrate with respect to x, use "rectangle − area under the curve", e.g. the area bounded by $y = e^x$, $y = 2$ and the y-axis = $2\cdot\ln 2 - \int_0^{\ln 2} e^x\,dx$ (2017F-Q5, 2019F-Q5).
5. Inverse-function area (2020A-Q11(e)): first recognise that the inverse of e^x is $g(x) = \ln x$ (1 mark on its own); then note that $\ln 2 \approx 0.69 < 1$, so the region is below the x-axis, and the area = $-\int_{\ln 2}^{1} \ln x\,dx$ (or written as $\int_{\ln 2}^{1} -\ln x\,dx$).
6. "Why are the two areas equal?": the two functions are moved by the same translation, so the shape is unchanged (2021F-Q5(b), two points: translated in the same direction and by the same distance, shape unchanged).

Mark allocation: ✓ intersection points/limits ✓ correct integral expression (integrand + limits + dx) ✓ antiderivative ✓ exact value.

Past questions: 2016A-Q13(d), 2016F-Q6, 2017F-Q5, 2018F-Q7(c), 2019F-Q5, 2020A-Q11(d)(e), 2021F-Q5, 2025F-Q5.

Pattern 6: Definite integral equations with an unknown parameter

Recognition: "If the area … is to equal 2, determine the exact value of k", "$\int_a^b 1/x\,dx = \ln 3$, find the relationship between a and b".

Approach: write down the integral → use the FTC to obtain an expression in terms of the parameter → set it equal to the given value → solve the equation (often using log laws: $\ln b - \ln a = \ln 3 \Rightarrow b = 3a$).

Past questions: 2017F-Q5(b), 2021F-Q7(a)(ii), 2020A-Q10(d).

Pattern 7: Riemann sums (rectangle estimates)

Recognition: Rectangles are drawn on the diagram, "demonstrate and explain why $L < \int < U$", "best estimate", "state one way to improve".

Approach

1. The lower sum uses the smaller function value on each subinterval (left endpoint or right endpoint, depending on whether the function is increasing or decreasing), and the upper sum uses the larger value; write out each expression separately.
2. Explanation: When the function is increasing on the interval, the left-endpoint rectangles are all below the curve, so it is an underestimate, and the right-endpoint rectangles are an overestimate; when the function is decreasing, the opposite is true (e.g. 1/x in 2021F-Q7). When the function is not monotonic on the interval, you cannot simply judge by endpoints. This explanation is worth 1 mark on its own (the 2017 report indicated that area-related explanation questions are generally weak).
3. Best estimate = (L + U)/2 (trapezium).
4. Ways to improve: increase the number of rectangles (reduce the width); use trapeziums; find the function expression and then use integration to calculate the exact value (the 2017F-Q9 key accepts the first and last).

Past questions: 2017F-Q9, 2021F-Q7(b), 2024F-Q6.

Pattern 8: Integral applications in context (cross-sectional area, volume, total change)

Recognition: Building or window cross-sections, trenches, dams, shampoo bottles; or "rate is … determine the total change".

Approach

1. Volume = cross-sectional area × thickness/length. Calculate the cross-sectional area using integration; units must be consistent (centimetres and metres).
2. Find intersection points or boundaries (e.g. h(W) = 0 to find the width, D(x) = −2 to find the trench edge).
3. Total change = $\int_a^b$ rate of change dt.
4. When the water level h is unknown: write the cross-sectional area as a function of h (the upper limit is an expression in h) → cross-sectional area × length (width) = volume → solve for h (2024A-Q16(b): width 4 cm). 2023A-Q14(a) uses the same method to derive the expression for V(h).

Marking points (2023A-Q14(a)): ✓ upper limit expressed in terms of h ✓ correct integral expression ✓ antiderivative ✓ substitute upper and lower limits ✓ simplify to the target expression.

Past questions: 2019A-Q15(c), 2020A-Q10(c)(d), 2022A-Q7, 2023A-Q9, 2023A-Q14, 2024A-Q16. (2021A-Q9 is the same kind of context, but the whole question does not involve integration.)